Linear Search

2 inputs
- ↳ array
- ↳ element

8	7	6	5
0	1	2	3

if search element = 5 ; it is worst case
if search element = 8 ; it is best case
if search element = 9 ; it is also worst case

Pseudocode

```
int search (int arr[], int x)
{
    n <- arr.length;                      ——  1
    for(int i=0; i<n; i++)                ——  1+n+1+n = 2+2n
    {
        if (arr[i] == x)                  ——  n
        {
            return i;                     ——  1
        }
    }
    // x is not found
    return -1;                            ——  1
}
```

True False

$1+2+2n+n+1+1$

$3n+5$

Ignore the Constant, Time Complexity is $\boxed{O(n)}$

Binary Search

Search for 25

a	2	7	9	11	20	25	27	50	51	60
	0	1	2	3	4	5	6	7	8	9
	start				middle					end

1. Define start and end
2. Find middle = $\frac{start + end}{2}$
3. Consider the floor value as mid.
4. If a[mid] < element, search right

start = middle + 1; end = end

25	27	50	51	60
5	6	7	8	9
start		mid		end

5. If a[mid] > element, search left side

start = start,
end = mid - 1

25	27
5	6
start mid	end

if start == mid, element found.

mid = $\frac{5+6}{2}$

$= \frac{11}{2}$

$= 5.5$

$\boxed{mid = 5}$

Pseudocode

```
int bin_search (cnt arr[], int x,
                int start, int end)
{
int mid = (start + end) / 2;
if (arr[mid] == x)
{
return mid;
}
if (start == end)
{
return -1;
}
if (arr[mid] > x)
{
return bin_search (arr, x, start, mid-1)
}
if (arr[mid] < x)
{
return bin_search (arr, x, mid+1, end);
}
}
```

Time Complexity

$T(n) = T\left(\frac{n}{2}\right) + c$ — 1

$T\left(\frac{n}{2}\right) = T\left(\frac{n}{4}\right) + c$ — 2

$T\left(\frac{n}{4}\right) = T\left(\frac{n}{8}\right) + c$ — 3

Apply Eqn 2 and 3 in eqn 1

$$T(n) = T\left(\frac{n}{2}\right) + c$$

$$= T\left(\frac{n}{4}\right) + c + c$$

$$= T\left(\frac{n}{4}\right) + 2c$$

$$= T\left(\frac{n}{8}\right) + 2c + c$$

$$= T\left(\frac{n}{8}\right) + 3c$$

Generalize

$$= T\left(\frac{n}{2^i}\right) + ic$$

$$= \frac{n}{2^i} = 1$$

$$n = 2^i$$

Apply log on both sides

$\log n = \log 2^i$

$\boxed{\log n = i}$

$= T(1) + \log n . c$

$= \cancel{1} + \cancel{c} . \log n$

$\boxed{O(\log n)}$